Adverse Drug Reaction Data Analysis

Safety signal detection in 500 spontaneous reports, 01 Jan 2026 to 28 Aug 2026 (India, fictional data)

Reports	Drugs / reactions	Serious	Hospitalisation	Unexpected
500	8 / 8	81 (16.2%)	35 (7.0%)	65 (13.0%)

1. Reporting frequency index

Definition. Reporting frequency index (RFI) = a drug-reaction pair's share of all reports divided by an equal share (1/8 = 12.5%). An RFI of 1.0 means the pair is reported as often as an average pair; above 1.0 it is reported more often. **Important feature of this dataset:** every drug is linked to exactly one reaction (and each reaction to one drug), so there are only 8 drug-reaction pairs and RFI per drug equals RFI per pair. No exposure (prescription) data is available, so RFI measures share of reports, not risk to patients.

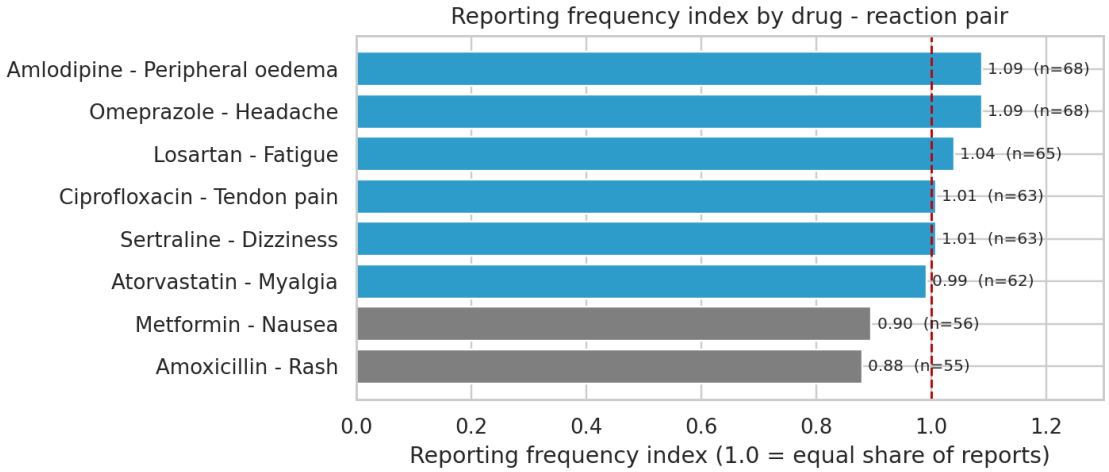


Figure 1. Reporting frequency index. Blue = 0.9 to 1.1 (about average), grey = below 0.9; no pair is above 1.1.

Drug	Reaction	Reports	Share	RFI
Omeprazole	Headache	68	13.6%	1.09
Amlodipine	Peripheral oedema	68	13.6%	1.09
Losartan	Fatigue	65	13.0%	1.04
Sertraline	Dizziness	63	12.6%	1.01
Ciprofloxacin	Tendon pain	63	12.6%	1.01
Atorvastatin	Myalgia	62	12.4%	0.99
Metformin	Nausea	56	11.2%	0.90
Amoxicillin	Rash	55	11.0%	0.88

The spread is narrow: RFI runs from 0.88 (Amoxicillin - Rash) to 1.09 (Amlodipine - Peripheral oedema and Omeprazole - Headache). With only 500 reports, differences of this size are within what chance alone could produce, so no pair stands out on frequency.

2. High-risk combinations: severity ratio

Definition. Severity ratio = serious-report rate of the pair divided by the serious-report rate of all other reports (overall serious rate 16.2%). A value above 1 means the pair's reports are more often serious. I flag **elevated** as a ratio of 1.25 or more. 95% confidence intervals use the log method; p-values are Fisher exact tests, shown raw and Bonferroni-adjusted for 8 comparisons.

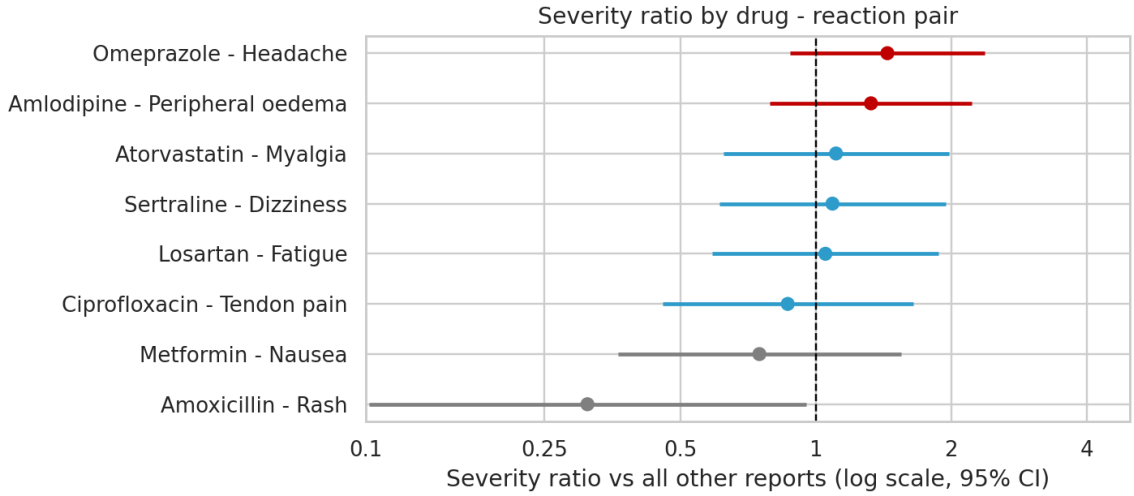


Figure 2. Severity ratio with 95% CI. Red = elevated (1.25 or more). Every interval crosses 1 except Amoxicillin - Rash, which is lower than average.

Drug - reaction	Reports	Serious	Rate	Severity ratio (95% CI)	p raw	p adj.	Hosp.	Unexp.
Omeprazole - Headache	68	15	22.1%	1.44 (0.88-2.38)	0.160	1.00	9	0%
Amlodipine - Peripheral oedema	68	14	20.6%	1.33 (0.79-2.22)	0.290	1.00	6	0%
Atorvastatin - Myalgia	62	11	17.7%	1.11 (0.62-1.98)	0.714	1.00	4	0%
Sertraline - Dizziness	63	11	17.5%	1.09 (0.61-1.94)	0.718	1.00	3	0%
Losartan - Fatigue	65	11	16.9%	1.05 (0.59-1.88)	0.857	1.00	6	100%
Ciprofloxacin - Tendon pain	63	9	14.3%	0.87 (0.46-1.65)	0.855	1.00	3	0%
Metformin - Nausea	56	7	12.5%	0.75 (0.36-1.55)	0.564	1.00	2	0%
Amoxicillin - Rash	55	3	5.5%	0.31 (0.10-0.95)	0.019	0.15	2	0%

Hosp. = reports with hospitalisation as the seriousness criterion. Unexp. = share of the pair's reports coded 'Unexpected'. Shaded rows = elevated severity ratio.

- **Elevated but not statistically supported:** Omeprazole - Headache (ratio 1.44, 15 of 68 serious, 9 hospitalisations) and Amlodipine - Peripheral oedema (ratio 1.33, 14 of 68 serious). Their confidence intervals include 1 and raw p-values are 0.16 and 0.29.
- **Lower than average:** Amoxicillin - Rash has the lowest severity (3 of 55 serious, ratio 0.31). Raw p = 0.019, but after correcting for 8 comparisons p = 0.15, so this is also not conclusive.
- **Across all pairs** serious rates do not differ significantly (chi-square p = 0.31).
- **Unexpected reactions:** the only pair coded 'Unexpected' is Losartan - Fatigue (all 65 reports); every other pair is coded 'Known'. In pharmacovigilance, unlabelled reactions get priority review even when severity is average (here 16.9% serious, 6 hospitalisations).

3. Reaction frequency across patient age brackets

Patients are aged 18 to 88 (median 53); there are no paediatric reports. Brackets: 18-29, 30-44, 45-59, 60-74, 75+.

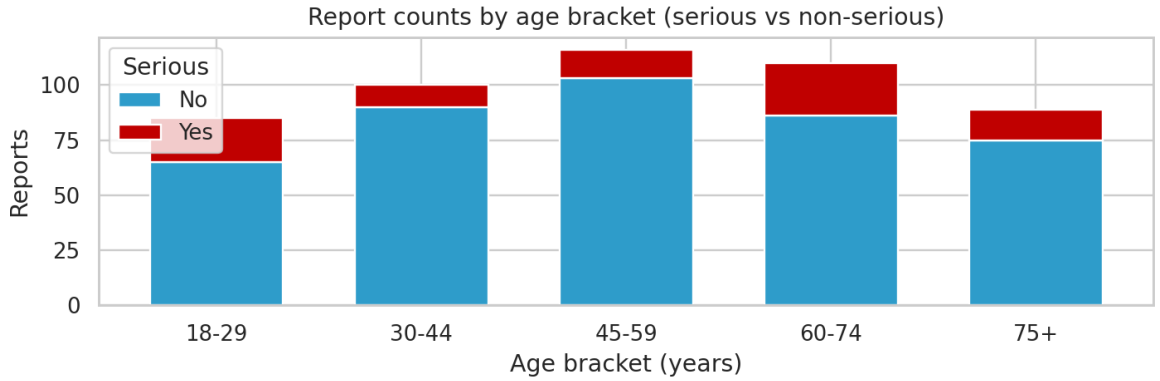


Figure 3. Number of reports by age bracket. Reporting is fairly even across ages (85 to 116 reports per bracket).

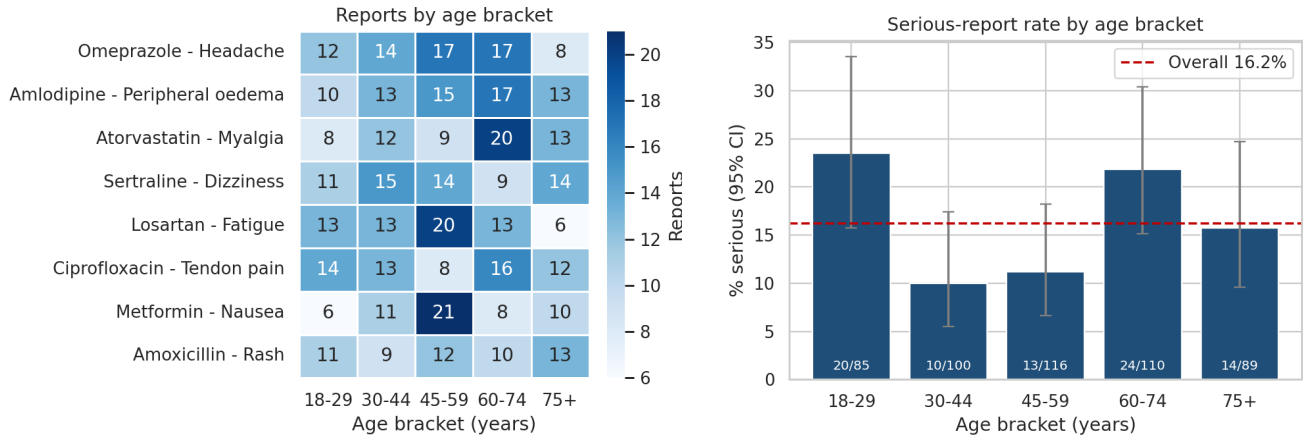


Figure 4. Left: reports per drug-reaction pair by age bracket. Right: serious-report rate by age bracket with 95% CI (dashed line = overall rate).

- **Reaction mix does not depend on age** (chi-square p = 0.35): no reaction is concentrated in a particular age group.
- **Seriousness does vary with age** (chi-square p = 0.028). The serious-report rate is 23.5% in 18-29 and 21.8% in 60-74, against 10.0% in 30-44 and 11.2% in 45-59; the 75+ group is 15.7%.
- This is one test among several in this report, with small counts (10 to 24 serious reports per bracket), so treat it as a lead to follow up, not a confirmed effect.

4. Safety signal detection summary

Signals are ranked by how much action they merit, combining severity, hospitalisations, expectedness and statistical support. None reaches statistical significance after correction for multiple comparisons.

Priority	Finding	Evidence	Suggested action
1 - Review	Losartan - Fatigue is the only pair coded 'Unexpected'	65 of 65 reports unexpected; 11 serious (16.9%), 6 hospitalisations	Check whether fatigue is in the reference label; case-level review of serious and hospitalised reports
2 - Monitor	Omeprazole - Headache has the highest severity ratio (1.44)	15/68 serious, 9 hospitalisations (most of any pair); CI 0.88-2.38, p = 0.16	Keep under routine monitoring; re-test as more reports arrive
3 - Monitor	Amlodipine - Peripheral oedema, severity ratio 1.33	14/68 serious, 6 hospitalisations; p = 0.29	Routine monitoring
4 - Lead	Higher seriousness in ages 18-29 and 60-74	24% and 22% serious vs about 10-11% in 30-59; p = 0.028 (unadjusted)	Check age-specific data in a larger dataset
No action	Amoxicillin - Rash is milder than average	3/55 serious; adj. p = 0.15	None; reassuring but not conclusive

Limitations and data quality

- **One reaction per drug.** Each drug is paired with a single reaction, so standard disproportionality measures such as the proportional reporting ratio or reporting odds ratio cannot be calculated (no comparison reports exist for a given reaction). Severity ratio and RFI were used instead.
- **No exposure data.** Without the number of patients taking each drug, report counts cannot be turned into incidence rates.
- **Dose and route look randomly assigned.** By my reading of standard adult dosing (not checked against a formulary), about 150 reports show clearly unusual doses (for example 500 mg of amlodipine or sertraline, or 5-20 mg of metformin or amoxicillin), and 160 list intravenous use for drugs normally given orally. I kept these records unchanged; they should be queried at source in real data.
- **Small counts.** Each pair has 55 to 68 reports and 3 to 15 serious ones, so confidence intervals are wide.
- **Spontaneous reporting** has under-reporting and reporting bias. These are fictional educational reports: frequency does not establish causation.

Method

Data checked for missing values and duplicate report IDs (none found; 500 unique reports, no blanks). Python (pandas, scipy, matplotlib, seaborn). Severity ratio confidence intervals: log method. Group comparisons: Fisher exact test (2x2) and chi-square (age tables). Multiple-comparison correction: Bonferroni across the 8 pairs.